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Ruminants

Identification of Cattle Using Nasolabial Plate Imprints and Biometric Analysis

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ABSTRACT

Background: Animal identification is a topic of many studies, with a range of biometric methods currently in use. The cattle muzzle serves as a unique source of biometric traits.

Objectives: The aim of this study was to determine the best method for muzzle visualisation using imprints, the most frequent forms and minutiae points on imprints, and the minimum number of minutiae points required to establish an identity profile.

Methods: Noseprints of 30 calves were taken on different surfaces and visualised using different methods (white paper and cardboard/ninhydrin solution and glass tile/small particle reagents and ceramic tile/fluorescent powder and glossy photopaper/grey instant or magnetic powder). The imprint of the entire muzzle was photographed and analysed using the Automated Fingerprint Identification System (AFIS) to detect the most frequent forms based on friction ridges and minutiae points. Further mathematical simulation revealed the minimal number of points required for animal identification.

Results: The best imprint was obtained on glossy black photopaper with grey instant powder. After analysing the digitised images with the AFIS magnifier, the six forms of beads and ridges in the selected 12 minutiae points were detected, thus creating an identity chart. Computer simulation confirmed that the lowest number of minutiae points necessary for unique animal identification, and 0% possibility of form repetition at points, was nine of the 12 selected points.

Conclusion: As a biometric method, the muzzle imprint in combination with AFIS has the potential to be stored on large scale and used internationally, enabling identity control that is not susceptible to the issues involved with other biometric methods.

1 | Introduction

Animal identification involves specifying all legal and physical traits that belong to a certain animal to distinguish it from other animals of the same species/breed (Mahmud et al. 2021; Zaripov et al. 2025). From the veterinary perspective, animal identification is one of the most important procedures in the profession. Proper identification is crucial in the assessment of animal health and welfare, controlling the transport of domestic and other animals and their products, especially for food chain control, import and transport, and the control of residues (Barron et al. 2009; Cihan et al. 2023; Fuentes et al. 2022; Hossain et al. 2022; Kumar et al. 2018).

In practice, there are several methods used for animal identification (Awad 2016; Cihan et al. 2023; Zaripov et al. 2025). However,

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in the light of growing animal welfare concern, the imperative is to choose methods that cause no pain, suffering, or permanent damage to the animal in equal or greater measure than caused by needlestick. The European Union (EU) legislation regulates animal identity based on ear tags and passports (Passantino 2013). These rules are also applied in the Republic of Croatia, as an EU Member State since 1 July 2013. However, in intensive breeding systems, reliable identification methods are difficult to achieve, particularly using adopted systems with ear tags. Awad (2016) described electronic marking as a more reliable biometric methods for animal identification, but also as a new technology applicable for animal tracking in achieving good agricultural and environmental practices. Despite this, non-biological methods such as puncture, tattooing, or stamping, can cause tissue damage or pain or stress in animals. In intensively kept animals, this can lead to health and production disturbances (Chan, Lin, Wang, et al. 2024; Chan, Lin, Ben, et al. 2024; Kumar et al. 2018). Furthermore, ear tags can change the physical appearance or behaviour of the animal (Awad 2016), endangering its welfare under intensive production conditions.

As Chan, Lin, Wang, et al. (2024) suggested, the use of biological identification methods is on the rise. Common biological identification methods, such as noseprints, facial images, retina scans, or other method based on an animal's description or photograph could be used as a reliable and permanent method for identification, without any harmful effects (Awad et al. 2013; Kumar et al. 2018). Based on the similarities between animals of the same species, apart from the determination of DNA profiles, differentiating anatomical features could also be used for identification. Therefore, similarities and differences between two or more animals of the same species could be determined by comparing specific traits. In recent decades, there has been a substantial number of studies regarding the use of biometric recognition technology for the identification and possible recognition of certain diseases (Chan, Lin, Ben, et al. 2024), though most have focused on cattle and dogs. In accordance with good veterinary practice, a method for identifying cows using a nasolabial plate imprint, called a noseprint, could be a fast, accurate, and inexpensive identification method that causes no harm or suffering to the animal.

Petersen (1922) described the smooth area of nose, or the nasolabial plate (*planum nasolabiale*), as a specific feature for cattle identification independent of sex, age, and breed. The nasolabial plate in cows, like the rostral plate in swine, is a modified, hairless, skin area at the top of the muzzle and upper lip, with large aggregations of eccrine nasolabial glands (*gldd. plani nasolabialis*), dispersed in subcutis that produce serous, almost watery, secretions that keep the surface moist at all times. The nasolabial plate is a modified skin layer covered with thick, moist, keratinised, stratified squamous epithelium, which is barely mobile and therefore has no role in food collection. The skin surface of the nasolabial plate has numerous slots forming specific islets, in which the glands secret their discharge, and their arrangement is specific to the individual. Therefore, due to their position and arrangement in lines, the nasolabial plate features unrepeatable, small polygonal areas that are unique to each individual. According to G. Li et al. (2022) and Zaripov et al. (2025), the small round-, oval-, or irregular-shaped protuberances on the nasal area are defined as beads, while the elongated grooves and valleys arranged in a particular manner are defined as ridges. Dyce et al. (1996) also mentioned this feature of unique nasolabial gland arrangement for animal identification. Furthermore, the muzzle pattern is a cattle dermatoglyphic trait equivalent to human fingerprints (G. Li et al. 2022). This unique pattern of glandular position on the nasolabial plate is of great research interest for the accurate identification of cows, compared to traditional methods (Kumar et al. 2018; Awad et al. 2013), giving the muzzle great biomarker characteristics (Ahmed et al. 2024). Muzzle identification is a relatively low-cost and simple method that is receiving increasing research interest, since the uneven and distinct features of the muzzle imprint enable unique cattle identification. As an imprint, the muzzle was studied, similar to human fingerprints, by Kumar et al. (2016). Further, Kumar et al. (2018) worked on cattle muzzle imprints to develop a deep learning-based approach and automated bovine identification system. Since then, numerous computer programs for screening this area have been described (Noviyanto and Arymurthy 2012; Sian et al. 2020; Zaripov et al. 2025; Zhang et al. 2023). The muzzle area has been studied and proposed in different notification systems for a number of animal species: dogs (Chan, Lin, Wang, et al. 2024), cats (Chen et al. 2016), pandas (Wang et al. 2019), and pigs (Chakraborty et al. 2020; Kujundžić et al. 2014). Other biometric approaches for accurate and reliable bovine identification include the use of animal side-view images (Hu et al. 2020). Artificial intelligence (AI)-based technologies that use face and head recognition have been described by Mahato and Neethirajan (2024). D. Li et al. (2024) described methods that combined cattle faces, muzzle patterns, and ear tags to avoid identification problems inherent in crowded farm conditions. The retina has also been studied, not only for identification purposes, but also as a source of disease detection, such as cardiovascular system in cattle (Cihan, Saygılı, Ermutlu, et al. 2024; Cihan et al. 2025). However, these studies used special software to process images and extract data, while the approach applied in the present paper is based on forensic methods for detection and identification.

The aim of this study was to identify cows based on their noseprints, with specific goals of determining (a) the best method for noseprint visualisation, (b) the most common beads and ridges on the animal identity chart identified by the noseprint, and (c) the minimum number of minutiae points required on each identity chart based on the selected bead and ridge positions to create a unique identity profile of the animal based on noseprint visualisation.

The key contributions of this study are as follows:

1. determination of the best method for cattle noseprint visualisation, based on four forensic methods for fingerprint and trace detection, since this first step in imprint visualisation is crucial. Based on our knowledge, this is the first study to apply forensic methods for trace detection in cattle imprint visualisation;
2. use Automated Fingerprint Identification System (AFIS) software for comparison and detection of beads and ridges at 12 minutiae points on each muzzle imprint, for the purpose of creating the animal's identity card;

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FIGURE 1 | Nasolabial area of cattle used for the imprint in this study (photo: M. Popović).

3. develop an animal identity card database based on the AFIS identification of special beads and ridges at 12 minutiae points for each animal, shared in Table 1 to encourage multidisciplinary studies;
4. develop computer analyses for differentiating animals based on identity charts created using AFIS, with high accuracy with the use of nine of 12 minutiae points, which could have future applications.

The study is organised as follows: Section 2 gives details on the collection of noseprints and data, processing and analyses; the results are presented in Section 3; the discussion is included in Section 4; and conclusions drawn from the study are summarised in Section 5. Appendix 1 gives the SAS 9.4 code created for the analyses.

2 | Material and Methods

2.1 | Animals

Noseprints were taken from 30 Simmental calves (15 male and 15 female), 3 months old (average body weight 150 kg), in the antechamber of the slaughterhouse MM MESNA INDUSTRIJA Ltd. (Donje Prekrižje, Krašić, Croatia), immediately before stunning. Independently of ear tags, calves were allocated the numbers 1–30.

2.2 | Noseprint Collection

Immediately before imprint, the nasolabial plate (Figure 1) was cleaned with a soft, moist towel. During noseprint collection, the imaging surface was firmly pressed against the nasolabial plate of the animal. Surfaces used for noseprint visualisation were 80 g/m^2 special A-4 white paper, glass tile, ceramic tile, cardboard, and black glossy photopaper (all by Bureau voor Dactyloscopische Artikelen, Netherlands). All collections were performed by the same person. Paper- and cardboard-based surfaces were held firmly against the palm, to obtain the same pressure and rigid surface for noseprint visualisation.

2.3 | Methods for Noseprint Visualisation Based on Surfaces Used

For noseprint visualisation, different reagents were used according to the surface used to take the noseprint. Noseprints on white paper or cardboard were visualised by suffusion of ninhydrin solution (2,2-dihidroxyindane-1.3-dion; Bureau voor Dactyloscopische Artikelen, Netherlands; Figure 2a). Noseprints taken on glass tile were visualised by sprinkling small particles reagents (SPR suspension; Bureau voor Dactyloscopische Artikelen, Netherlands), while those taken on ceramic tiles were visualised in a UV chamber (Bureau voor Dactyloscopische Artikelen, Netherlands) after sprinkling with fluorescent powder (Bureau voor Dactyloscopische Artikelen, Netherlands; Figure 2b). Noseprints taken on black glossy photopaper were visualised with grey instant (Figure 3a) or magnetic powder (Figure 3b) (Bureau voor Dactyloscopische Artikelen, Netherlands). After visualisation, imprints were photographed (Canon EOS 400d, objective Canon macro 50 mm, f/2.5) for a permanent record representing the identity chart, or identification of the animal.

2.4 | Methods for Creation of a Unique Identity Profile of the Animal Based on the Noseprint

After obtaining the animal identity chart of the highest quality, depending on the imaging surface used, specific beads and ridges were analysed for animal identification with the AFIS magnifier (Automated Fingerprint Identification System, Safran MORPHO META MORPHO 3.2). The AFIS system selected six specific forms detected in the highest frequency in all analysed noseprints, similar to minutiae in human finger print authentication (Maltoni et al. 2009; Pukšić and Žagar 2016) (Figure 4).

Based on the AFIS analysis, the specific glandular arrangement enabled the identification of 12 minutiae points for each examined identity chart. The specific features at each point were classified in one of six distinctive beads and ridges (Figures 2–3), thus forming a unique identity profile of the animal (Figures 5–6) based on its noseprint.

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TABLE 1 | Identity chart, with number of forms by animal and minutiae point.

Animal ID	Minutiae point												Total number of forms recorded per ID					
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	A	B	C	D	E	F
1	E	A	E	B	A	A	E	A	C	B	E	F	4	2	1	0	4	1
2	F	A	E	F	E	E	E	E	A	E	E	E	2	0	0	0	8	2
3	A	C	C	E	E	E	A	F	D	D	E	F	2	0	2	2	4	2
4	A	F	F	D	B	B	C	A	A	A	C	C	4	2	3	1	0	2
5	B	B	C	C	D	A	A	A	B	E	E	F	3	3	2	1	2	1
6	C	E	F	E	F	A	A	A	C	C	D	D	3	0	3	2	2	2
7	A	B	B	D	D	E	E	F	E	E	A	A	3	2	0	2	4	1
8	A	A	C	C	D	D	A	A	A	F	F	E	5	0	2	2	1	2
9	E	E	F	F	C	D	D	E	E	A	A	B	2	1	1	2	4	2
10	F	F	F	A	A	C	C	D	D	E	F	E	2	0	2	2	2	4
11	F	B	B	A	B	C	A	A	E	D	D	F	3	3	1	2	1	2
12	A	F	F	D	F	F	B	F	D	D	A	A	3	1	0	3	0	5
13	A	C	C	D	C	F	E	E	D	A	B	B	2	2	3	2	2	1
14	B	F	A	A	C	D	D	E	E	F	F	A	3	1	1	2	2	3
15	B	B	B	C	E	E	D	D	D	A	B	C	1	4	2	3	2	0
16	B	C	C	D	C	C	E	F	F	A	A	A	3	1	4	1	1	2
17	C	D	A	A	B	B	D	E	D	D	F	E	2	2	1	4	2	1
18	A	E	E	F	F	D	D	A	A	C	C	A	4	0	2	2	2	2
19	C	E	F	F	A	A	D	C	C	B	B	F	2	2	3	1	1	3
20	A	D	E	B	C	A	C	C	D	D	E	E	2	1	3	3	3	0
21	E	C	D	D	E	E	F	E	E	F	F	B	0	1	1	2	5	3
22	F	A	A	B	E	E	F	C	C	C	E	E	2	1	3	0	4	2
23	F	F	D	B	B	B	E	C	A	A	A	D	3	3	1	2	1	2
24	D	D	B	B	E	F	E	B	B	A	B	A	2	5	0	2	2	1
25	A	C	C	F	F	E	E	D	C	C	C	D	1	0	5	2	2	2
26	D	E	E	F	F	D	D	A	A	B	A	A	4	1	0	3	2	2
27	E	A	C	C	C	D	D	E	F	F	D	D	1	0	3	4	2	2
28	E	D	D	E	D	C	A	A	A	A	B	A	5	1	1	3	2	0
29	B	C	C	A	A	C	B	B	A	F	F	F	3	3	3	0	0	3
30	F	D	A	A	B	C	E	E	C	C	F	C	2	1	4	1	2	2
Total number of forms recorded per point																		
A	9	5	4	6	4	5	6	9	8	8	6	8	78					
B	5	4	4	5	5	3	2	2	2	3	5	3		43				
C	3	6	8	4	6	6	3	4	6	5	3	3			57			
D	2	5	3	6	4	6	8	3	7	5	3	4				56		
E	5	5	5	3	6	7	9	8	5	4	6	6					69	
F	6	5	6	6	5	3	2	4	2	5	7	6						57

Note: Distribution and number of selected forms of beads and ridges based on the nasolabial gland arrangement of the nasolabial plate on 30 identity charts of calf noseprints (N = 30) at 12 minutiae points: line (A), fork (B), hook (C), hook combined with fork (D), full islet (E), and end of line and in the same plane the continuation of the line with interruption (F).

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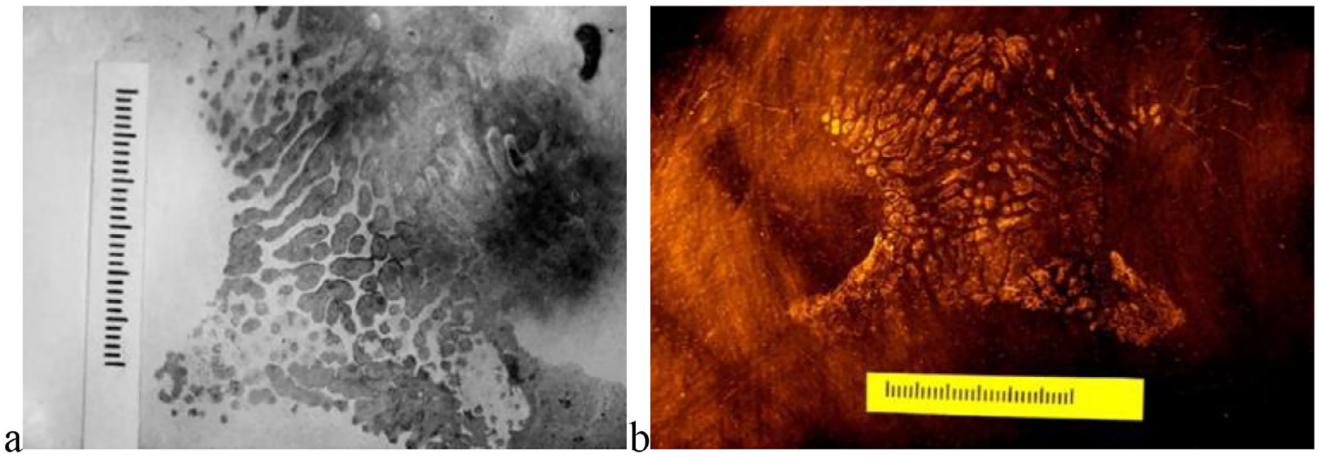


FIGURE 2 | Identity chart of calf noseprints visualised by (a) cardboard and ninhydrin solution and (b) ceramic tile and fluorescent powder.

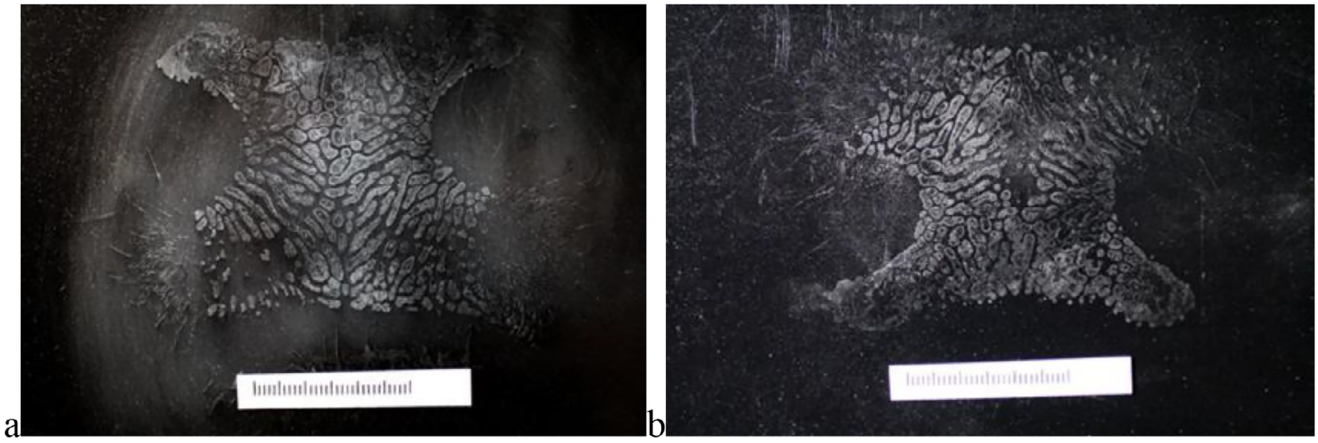


FIGURE 3 | Identity chart of noseprints visualised using black glossy photopaper and (a) grey instant or (b) magnetic powder.

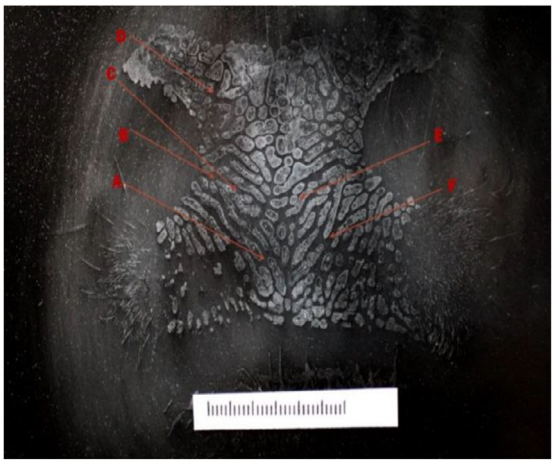


FIGURE 4 | Six specific forms of beads and ridges identified in the nasolabial gland arrangement on the nasolabial plate of the identity chart of a calf, whose noseprint was visualised with black glossy photopaper and grey instant powder: line (A), fork (B), hook (C), hook combined with fork (D), full islet (E), and end of line and in the same plane the continuation of the line with interruption (F).

2.5 | Calculation

This study was conducted on a computer equipped with an AMD Ryzen 5 3600 6-Core processor and 16 GB RAM. Statistical data analysis was performed using the SAS program language within the statistical program SAS 9.4 (Statistical Analysis Software 2002–2012, SAS Institute Inc., Cary, USA). Descriptive statistics were performed using SAS modules PROC MEANS, PROC UNIVARIATE, and PROC FREQ. For the purposes of this study, a macro was created in SAS 9.4 (Appendix 1), into which all data were entered and edited. Using the SURVEYSELECT procedure, the number of minutiae points for which we wanted to examine matches between animals was randomly selected from 12 possible points. For each number of points, we started random replication 100 times to obtain the proportion, mean, 95% confidence interval, and minimum and maximum number of matches. We determined the theoretical probability of matches for a certain number of points using the binomial distribution, for which we assumed that the forms (A–F) are equally distributed, equally probable, and independent between the points. Observed at one point, the probability for a particular form is $p = 1/6$. If k is the number of points in which they match and if the points are mutually independent, then k has a binomial distribution with parameters p and n, where we repeat n independent trials in

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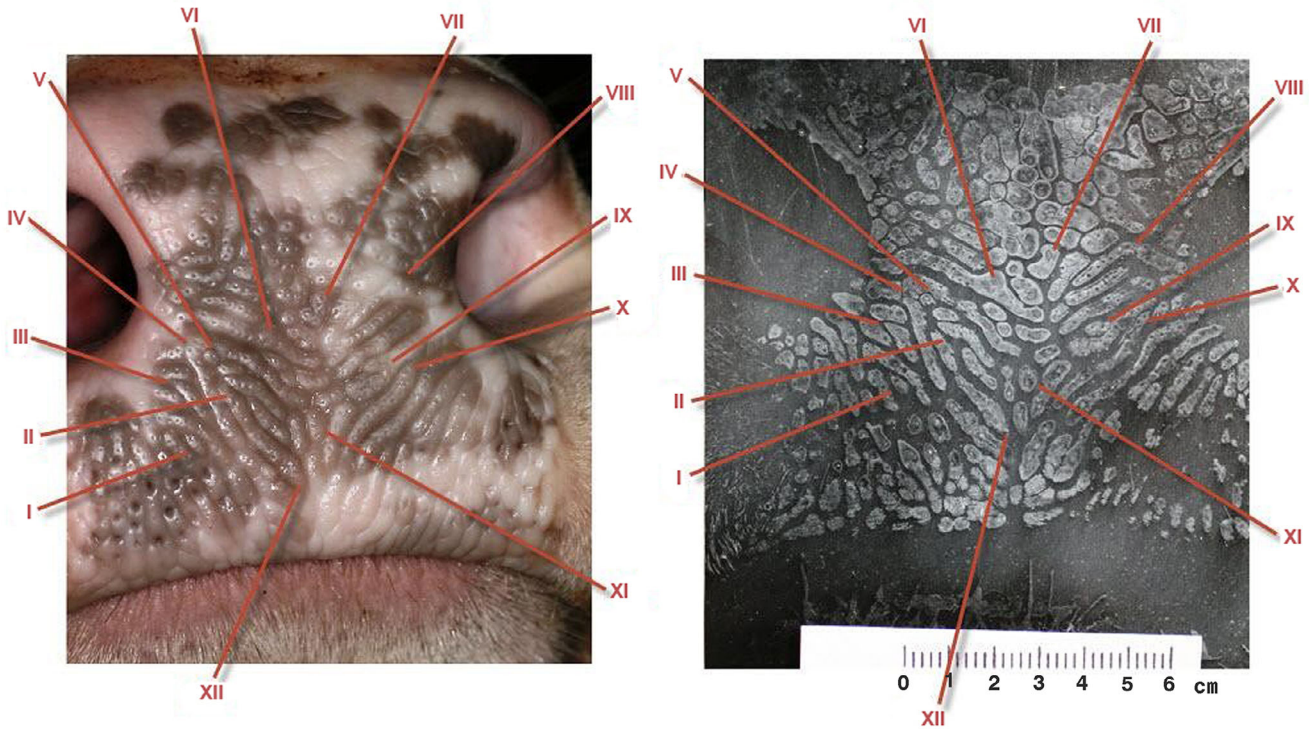


FIGURE 5 | Identification profile of calf ID 1 in 12 minutiae points, clockwise: I. full islet (E), II. line (A), III. full islet (E), IV. fork (B), V. line (A), VI. line (A), VII. full islet (E), VIII. line (A), IX. hook (C), X. fork (B), XI. full islet (E), and XII. end of line and in the same plane the continuation of the line with interruption (F).

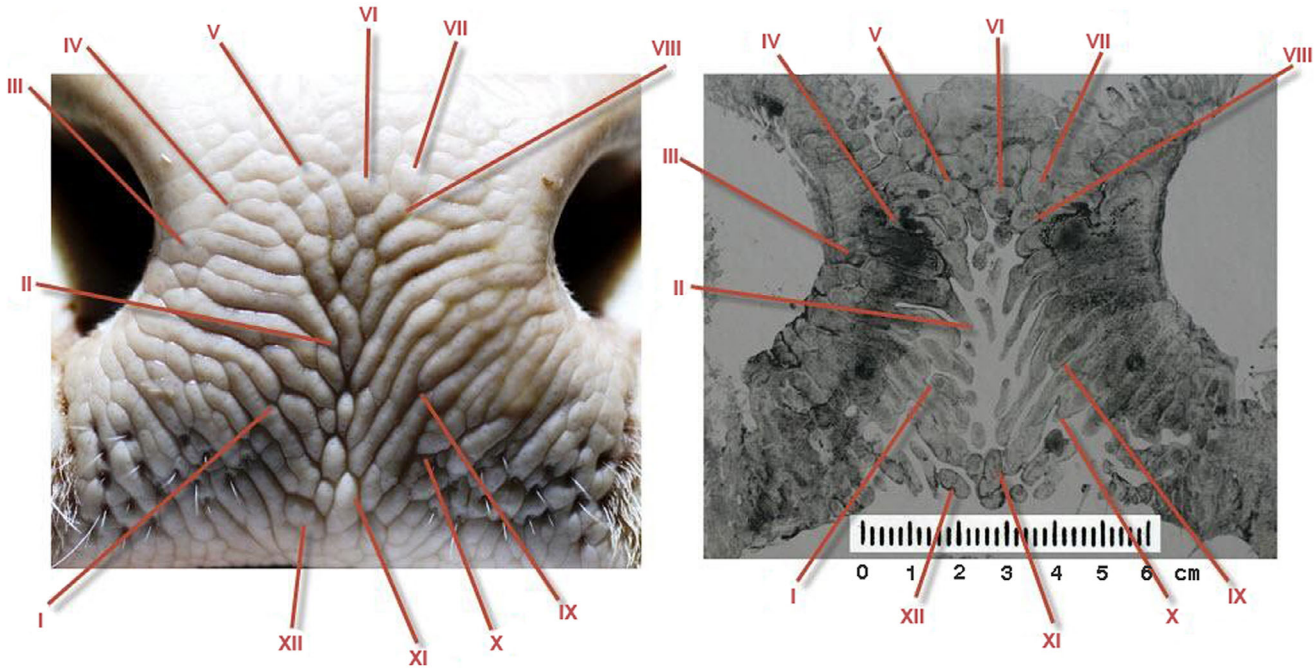


FIGURE 6 | Identification profile of calf ID 2 in 12 minutiae points, clockwise: I. end of line and in the same plane the continuation of the line with interruption (F), II. line (A), III. full islet (E), 4. end of line and in the same plane the continuation of the line with interruption (F), 5. full islet (E), 6. full islet (E), 7. full islet (E), 8. full islet (E), 9. line (A), 10. full islet (E), and 11. full islet (E), 12. full islet (E).

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which the probability of success (matching) is the same in each (p), and the probability of failure is $q = 1-p$ and our expression can be written as:

$$\text{Probability} = \frac{n!}{k! \times (n-k)!} \times \left(\frac{1}{6}\right)^k \times \left(\frac{5}{6}\right)^{12-k}. \quad (1)$$

We verified this formula by simulating a million animals with 12 minutiae points and six forms per point and obtained a match between theoretical distributions and posterior probabilities.

The time range needed for the study differed, with noseprint collection being very brief at up to 5 s per animal, and easily obtainable in the slaughterhouse. Visualisation took up to 30 min per imprint. Once the best imprints were detected, AFIS analyses took several minutes per imprint. Reading shapes and forms at the points in the AFIS analysed imprints, thus creating the database, took ~3 h. Once the simulation program in SAS 9.4 was created, the time needed for this simulation was up to 5 s.

3 | Results

The obtained fragments of the imprint (80 g/m^2 special A-4 white paper, cardboard) or partially obtained imprint (up to 60% with the use of glass or ceramic tile) of the nasolabial plate were of insufficient quality to be used for animal identification (Figure 2).

The methods using black glossy photo paper for the nasolabial plate imprint, and visualisation with grey instant or magnetic powder resulted in identity charts of the highest quality, with the finest contrast and detailed forms of beads and ridges necessary for noseprint identification (Figure 3).

The AFIS magnifier was used to examine 60 identity cards (two per animal) obtained with black glossy photopaper and visualised with grey instant (one card per animal, 30 cards in total) or magnetic powder (one card per animal, 30 cards in total). Six specific forms of beads and ridges were detected in the friction ridges formed on the nasolabial plate, which had the highest frequency of visualisation on the examined charts, independent of their position on the nasolabial plate. They were named according to human minutiae in the friction ridges on fingerprints (Maltoni et al. 2009; Pukšić and Žagar 2016). Figure 4 shows the selected forms on the noseprint: line (A), fork (B), hook (C), hook combined with fork (D), full islet (E), end of line and in the same plane the continuation of the line with interruption (F). Their frequency is shown in Table 1. Since the six selected forms were of the best quality on identity charts of noseprints obtained with black glossy photopaper and visualised with grey instant powder, only these identity charts (one per animal, 30 cards in total) were used for the selection of the minimum number of minutiae points. These 30 cards were examined with the AFIS magnifier in a clockwise direction, starting at the lower left corner and moving up and then down toward the lower right corner in a semi-circular direction. Therefore, 12 minutiae points (marked in roman numerals) were taken as the minimum number of points needed to create the unique identity profile of the animal (Table 1).

Table 1 presents the distribution of forms per animal and minutiae point. The table also shows that the most common form (of the six selected forms) was the line (A), followed by the full islet (E). Phenotype shapes, such as the hook (C), hook combined with fork (D), and end of line and in the same plane the continuation of the line with interruption (F), had the same occurrence, while the least common form was the fork (B). Form A was most common at minutiae points I, VIII, IX, X, and XII. Form E was most numerous at points VI and VII, form C at points II and III, and form F at point XI. Point IV had an equal occurrence of forms A, D, and F, while point V had an equal frequency of forms C and E. Though form A was most common overall and for the majority of the selected minutiae points (5 of 12 points), the difference in frequency between the points was not statistically significant.

Figure 5 shows the identification profile of calf ID 1 at 12 minutiae points on the nasolabial plate, clockwise: I. full islet (E), II. the end of line (A), III. full islet (E), IV. fork (B), V. line (A), VI. line (A), VII. full islet (E), VIII. line (A), IX. hook (C), X. fork (B), XI. full islet, XII. end of line and in the same plane the continuation of the line with interruption (F). Therefore, the identity card of calf ID 1 shows the presence of five of six of the selected forms at the 12 minutiae points: 4 A, 2 B, 1 C, 4 E, and 1 F.

However, on the identification card of calf ID 2 (Figure 6) in the 12 minutiae points, only three of the six selected forms were found (2 A, 8 E, and 2 F), making its identity profile completely different from calf ID 1. Based on these identity profiles, it is clear that these two calves differ in 75% (nine of 12) of points, overlapping only in points II, III, and XI. Therefore, we calculated the minimum number of randomly selected points required for animal identification, that is, with no repetitions in any other identity.

Table 2 shows the frequency and percentage of total number of matches per minutiae point.

The lowest frequency of matches was found for point II, with the highest frequency for point VII. Based on mathematical formula (1), the probability of animals matching in any selected number of given points decreases as the number of points increases (Table 3). The matching probability for five points is less than 0.05 (0.02844), and the matching probability for eight points is less than 0.001 (0.00014).

Table 4 shows the descriptive statistics of the dataset, that is, matches in any number of given minutiae points and by animal ID. The highest percentage of matches was in two points (26.9% of matches), with a decreasing trend up to six points (0.67% of matches). Matches in seven and eight points were 0%, in nine points 0.20%, and then dropping to 0% for all remaining numbers of points. Based on the mathematical formula and the results obtained, we concluded that nine minutiae points are sufficient for animal identification and conducted a control through computer simulation (Table 5). When 11 or 10 points were randomly chosen, in 100 replications, the matching in 10 or 11 points occurred in 0% of cases, leaving nine points as the optimal choice. Once the number of randomly chosen points was nine or fewer, the percentage of matches in these points was always above 0%.

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TABLE 2 | Frequency and percentage of total number of matches per minutiae point.

Total number of matches	Minutiae point												Total
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	
N	150	122	136	128	124	134	168	160	152	134	134	140	1682
Percent	8.92	7.25	8.09	7.61	7.37	7.97	9.99	9.51	9.04	7.97	7.97	8.32	100.0

TABLE 3 | Probability of animals matching in none, one or more minutiae points calculated by a mathematical formula.

Matches in number of points (*k*)	Probability
0	0.112156654784620
1	0.269175971483076
2	0.296093568631383
3	0.197395712420922
4	0.088828070589415
5	0.028424982588613*
6	0.006632495937343*
7	0.001136999303545*
8	0.000142124912943**
9	0.000012633325595**
10	0.000000757999536**
11	0.000000027563619**
12	0.000000000459394**

Probability of animals matching in none, one or more points calculated by a mathematical formula (1), where $n = 12$.

*Matching probability $p < 0.05$; **matching probability $p < 0.001$.

4 | Discussion

In nature, every animal differs from every another animal, whereby its identity, depending on the species and breed, can be easily determined. However, animals of the same species/breed are very similar and cannot be distinguished without the use of certain methods based on differentiating traits to perform identification. Thus, identity is the totality of the invariable characteristics that make up a particular animal, according to which it can be distinguished from all others (Merck 2012). Specifically, to identify means to compare the identity of an unknown animal to previously known animals, based on certain identifying features. Traits used in the identification process must have certain features: universality (possessed by every individual of that species), individuality (differs among individuals of the same species), durability, and immutability (Awad et al. 2013), accompanied by achievable performance and circumvention of possible fraudulent activities (Cihan, Saygılı, Akyüzlü, et al. 2024). Furthermore, these traits must have the ability to be singled out from the totality of the animal's physical appearance (for database creation and comparison) and must be easily collected and used for identification purposes. One such trait in animals are the forms of beads and ridges formed by the nasolabial gland arrangement on the muzzle, that is, nasolabial plate in cattle (Dyce et al. 1996; Nickel et al. 1987).

Animal identification based on muzzle print images is already known (Ahmed et al. 2024; Bello et al. 2020; Kumar et al. 2018; Sian et al. 2020; Zaripov et al. 2025). However, previous studies have used special software to process muzzle images and extract data. The present approach differs in treating the muzzle as a forensic trace and obtaining the imprint using forensic methods. Therefore, different surfaces were used, giving varying results in the quality of obtained imprint. For the reliability of identification based on a biometric approach, the quality of the imprint image is a critical factor (Bouhamed 2025), and the entire surface needs to be captured, with good visualisation of ridges. The best results in this study were obtained with the use of glossy photopaper and grey instant powder. It is possible that the moisture of the area requires a dark glossy surface for a better imprint. On the other hand, obtaining and visualising the muzzle imprint is a time- and labour-consuming process. The calm nature of bovine species makes it possible to obtain nose print visualisation by firming pressing the surface across the whole muzzle area. Cattle are accustomed to human contact, and the speed of the process makes it possible to use in the cow's daily environment, unlike dogs, which are sensitive and restless regardless of human contact, making noseprint visualisation more demanding (Chan et al. 2024b). The imprint is easily taken, thus avoiding the need for repeated photography and image processing, such as in methods that use the image of anatomic traits, either nose or retina in cattle (Ahmed et al. 2024; Bello et al. 2020; Cihan, Saygılı, Akyüzlü, et al. 2024; Cihan et al. 2025; Kumar et al. 2018; Sian et al. 2020; Zaripov et al. 2025), or dogs (Chan, Lin, Wang, et al. 2024).

Hence, this is the first time that cattle identification has been based on the distinctive forms of beads and ridges of the nasolabial plane, on the noseprint taken on black glossy photopaper and visualised with and grey instant powder. Like the papillary lines in humans (Maltoni et al. 2009; Pukšić and Žagar 2016), the patterns of beads and ridges of the nasolabial plane are unique to each cow, making the unique imprint suitable for animal identification, as described herein. This kind of identification is based also on the theory of probability (Barry et al. 2007), by determining the specific overlap of two or more noseprints in the characteristic morphological shapes formed by the nasolabial glands, and which are examined here in a similar manner as papillary patterns (Noviyanto and Arymurthy 2013). Based on the basic papillary shapes (arch, loop, whorl) (Moses et al. 2011), six forms were selected from cattle noseprints and defined as: line (A), fork (B), hook (C), hook combined with fork (D), full islet (E), end of line, and in the same plane the continuation of the line with interruption (F). However, unlike several methods

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TABLE 4 | Descriptive data of matches by number of minutiae points and by animal ID.

Animal ID	Number of animals with matches in none, one or more minutiae points													Total matches	
	0	1	2	3	4	5	6	7	8	9	10	11	12	N	% †
1	4	7	11	2	4	1	0	0	0	0	0	0	0	25	86.2
2	2	7	7	11	1	0	1	0	0	0	0	0	0	27	93.1
3	5	6	4	9	5	0	0	0	0	0	0	0	0	24	82.8
4	2	9	11	4	2	1	0	0	0	0	0	0	0	27	93.1
5	6	9	5	5	3	1	0	0	0	0	0	0	0	23	79.3
6	9	4	8	5	2	1	0	0	0	0	0	0	0	20	69.0
7	5	5	8	9	0	2	0	0	0	0	0	0	0	24	82.8
8	2	6	11	5	3	2	0	0	0	0	0	0	0	27	93.1
9	4	8	9	1	4	3	0	0	0	0	0	0	0	25	86.2
10	4	11	5	6	3	0	0	0	0	0	0	0	0	25	86.2
11	2	12	6	7	2	0	0	0	0	0	0	0	0	27	93.1
12	6	7	7	4	4	1	0	0	0	0	0	0	0	23	79.3
13	4	5	8	5	6	0	1	0	0	0	0	0	0	25	86.2
14	4	9	5	5	3	3	0	0	0	0	0	0	0	25	86.2
15	2	8	14	4	1	0	0	0	0	0	0	0	0	27	93.1
16	4	7	6	8	2	1	1	0	0	0	0	0	0	25	86.2
17	5	5	14	1	2	1	1	0	0	0	0	0	0	24	82.8
18	2	12	4	5	4	1	0	0	0	1	0	0	0	27	93.1
19	4	12	8	1	2	2	0	0	0	0	0	0	0	25	86.2
20	2	12	7	4	4	0	0	0	0	0	0	0	0	27	93.1
21	6	7	7	6	3	0	0	0	0	0	0	0	0	23	79.3
22	8	6	7	4	3	0	1	0	0	0	0	0	0	21	72.4
23	2	9	11	6	0	1	0	0	0	0	0	0	0	27	93.1
24	4	12	7	3	3	0	0	0	0	0	0	0	0	25	86.2
25	4	5	11	5	3	1	0	0	0	0	0	0	0	25	86.2
26	6	6	7	7	1	1	0	0	0	1	0	0	0	23	79.3
27	5	8	10	3	0	3	0	0	0	0	0	0	0	24	82.8
28	2	8	9	8	2	0	0	0	0	0	0	0	0	27	93.1
29	5	8	7	5	4	0	0	0	0	0	0	0	0	24	82.8
30	4	8	8	4	4	0	1	0	0	0	0	0	0	25	86.2
Total															
N	124	238	242	152	80	26	6	0	0	2	0	0	0	/	/
Pairs/No. of duplications															
N	/	119	121	76	40	13	3	0	0	1	0	0	0	/	/
Mean	4.13	7.93	8.06	5.06	2.66	0.86	0.20	0	0	0.06	0	0	0	/	/
SD	1.83	2.39	2.61	2.40	1.47	0.97	0.40	0	0	0.25	0	0	0	/	/
Min—max	2-9	4-12	4-14	1-11	0-6	0-3	0-1	0	0	0-1	0	0	0	/	/
Percent															
% ‡	13.8	26.4	26.9	16.9	8.87	2.87	0.67	0.00	0.00	0.20	0.00	0.00	0.00	/	/

Abbreviation: SD, standard deviation.

% †—Percentage of animal matches in all given minutiae points with any other animal (total matches by animal (N) / 29 × 100).

% ‡—Probability of animals matches in none, one or more minutiae points (mean value of each point (mean) / 30 × 100).

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TABLE 5 | Computer simulation and descriptive statistics of matches in a randomly chosen number out of 12 minutiae points (100 replications).

Randomly chosen number of individual points (100 replications)	Matches in the number of points	Number of animal pairs matching in one or more points in 100 replications						
		Mean	Max	Min	SD	UCLM	LCLM	% ‡
11	1	129.21	134	119	3.47	129.90	128.52	28.713
	2	119.24	129	111	3.70	119.97	118.51	26.498
	3	70.46	83	59	5.26	71.50	69.42	15.658
	4	32.01	40	26	3.74	32.75	31.27	7.113
	5	9.28	12	6	1.67	9.61	8.95	2.062
	6	1.44	3	0	0.82	1.60	1.28	0.320
	7	0	0	0	0.00	0.00	0.00	0.000
	8	0.72	1	0	0.45	0.81	0.63	0.160
	9	0.28	1	0	0.45	0.37	0.19	0.062
	10	0	0	0	0	0	0	0.000
	11	0	0	0	0	0	0	0.000
10	1	139.48	152	125	5.36	140.54	138.42	30.996
	2	117.53	129	104	5.72	118.66	116.40	26.118
	3	62.83	80	51	5.73	63.97	61.69	13.962
	4	24.4	32	16	3.81	25.16	23.64	5.422
	5	5.93	10	2	1.82	6.29	5.57	1.318
	6	0.69	2	0	0.66	0.82	0.56	0.153
	7	0.47	1	0	0.50	0.57	0.37	0.104
	8	0.47	1	0	0.50	0.57	0.37	0.104
	9	0.06	1	0	0.24	0.11	0.01	0.013
	10	0	0	0	0	0	0	0.000
9	1	148.63	163	134	6.84	149.99	147.27	33.029
	2	113.52	128	101	5.60	114.63	112.41	25.227
	3	52.91	65	39	5.20	53.94	51.88	11.758
	4	17.36	25	9	3.28	18.01	16.71	3.858
	5	3.35	7	1	1.40	3.63	3.07	0.744
	6	0.63	2	0	0.66	0.76	0.50	0.140
	7	0.51	1	0	0.50	0.61	0.41	0.113
	8	0.13	1	0	0.34	0.20	0.06	0.029
	9	0.01	1	0	0.10	0.03	−0.01	0.002
8	1	157.63	177	140	7.77	159.17	156.09	35.029
	2	105.72	119	88	6.96	107.10	104.34	23.493
	3	43.69	56	35	4.31	44.55	42.83	9.709
	4	12.01	18	6	2.65	12.53	11.49	2.669
	5	1.81	6	0	1.24	2.06	1.56	0.402
	6	0.5	3	0	0.64	0.63	0.37	0.111
	7	0.19	1	0	0.39	0.27	0.11	0.042
	8	0.03	1	0	0.17	0.06	0.00	0.007
7	1	163.26	183	140	9.77	165.20	161.32	36.280
	2	95.89	112	76	6.70	97.22	94.56	21.309
	3	32.86	42	23	4.21	33.70	32.02	7.302

(Continues)

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TABLE 5 | (Continued)

Randomly chosen number of individual points (100 replications)	Matches in the number of points	Number of animal pairs matching in one or more points in 100 replications						
		Mean	Max	Min	SD	UCLM	LCLM	% ‡
	4	7.11	12	0	2.09	7.52	6.70	1.580
	5	1	5	0	0.93	1.18	0.82	0.222
	6	0.36	2	0	0.50	0.46	0.26	0.080
	7	0.03	1	0	0.17	0.06	0.00	0.007
6	1	169.37	193	147	9.94	171.34	167.40	37.638
	2	82.12	101	64	7.48	83.60	80.64	18.249
	3	23.31	32	15	3.65	24.03	22.59	5.180
	4	3.67	7	1	1.63	3.99	3.35	0.816
	5	0.54	3	0	0.76	0.69	0.39	0.120
	6	0.11	1	0	0.31	0.17	0.05	0.024
5	1	165.56	186	139	10.55	167.65	163.47	36.791
	2	66.69	82	47	6.28	67.94	65.44	14.820
	3	14.67	23	7	3.56	15.38	13.96	3.260
	4	1.78	6	0	1.28	2.03	1.53	0.396
	5	0.19	1	0	0.39	0.27	0.11	0.042
4	1	160.68	191	131	12.91	163.24	158.12	35.707
	2	47.54	60	33	6.24	48.78	46.30	10.564
	3	7.55	12	2	2.41	8.03	7.07	1.678
	4	0.59	3	0	0.65	0.72	0.46	0.131

Abbreviations: LCLM, lower confidence limit; SD, standard deviation; UCLM, upper confidence limit.

% †— Percentage of animal matching in all given points with any other animal (total matchings by animal (N) / 29 × 100).

% ‡—Probability of animals matching in one or more points ((mean/15) / (30)) × 100.

used for fingerprint visualisation in humans (Maltoni et al. 2009; Stolić et al. 2019), due to the high physiological moisture and anatomical curvature of the nasolabial plate in cows, the tested method with black glossy photopaper and grey instant powder proved to be the only adequate method for noseprint visualisation in this study. Comparison of traces and imprints in at least 12 minutiae points by AFIS, according to Mršić et al. (2014), was successful in identifying calves with high accuracy, although on a small sample.

However, this method differs from methods used for the muzzle imprint of sheep, goats, pigs, dogs, and cats, for the creation of a unique identification profile (Kujundžić et al. 2014; Pajurin et al. 2018). Apart from the use of AFIS, our method does not require the use of highly sophisticated computer hardware or software or a deep learning base approach, making it more usable and applicable. Furthermore, after the imprint is taken and digitised, a DNA molecule can be completely isolated and analysed from the imprint of the cattle nasolabial plate (Modly and Mršić 2014). Therefore, this biometric identification system could be used in conjunction with DNA methods of animal identification (Primorac and Schanfield 2023). Unlike Minagawa et al. (2002), the cattle noseprint identification method is non-invasive and painless and does not change the appearance of the animal. Consequently, it does not affect the behaviour or survival of the animal nor does it affect its welfare in intensive farming conditions. Similar to retinal biometric images, the noseprint has the advantage of being applicable to animals of all ages and sizes, and it is unique to the animal and does not change over time. However, unlike retinal images, it is less costly, and it is not subject to error due to external influences such as lightning or animal restlessness (Saygılı et al. 2024). It does not require powerful, expensive, or sophisticated hardware and software, which can be costly to acquire and maintain (Cihan et al. 2023). The noseprint method is inexpensive, durable, and easily stored in digital form, like any friction ridge record (Hall 2023).

Consequently, the digitised noseprint of cattle would enable procedures to determine cow identity within the framework of contemporary intensive breeding, based on AFIS biometric technology, allowing rapid search of databases and linking between two impressions (Gibb and Riemen 2023). Although numerous computer programs for the screening of noseprints (Noviyanto and Arymurthy 2012; Sian et al. 2020; Zaripov et al. 2025; Zhang et al. 2023) and AI face (D. Li et al. 2024; Mahato and Neethirajan 2024) and head with ear tags (Pretto et al. 2024; Smink et al. 2024) recognition technology are being developed, the results presented here are an inspiration for the creation of a new automated system for cow identification using noseprints based on the unique patterns of beads and ridges in the nasolabial plate (A, B, C, D, E,

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F), similar to automated fingerprint identification system—AFIS (Komarinski 2004). The widespread use of AFIS systems makes this approach interesting and easily implementable in everyday animal work. Namely, in farmed cattle breeding conditions with the help of AFIS's similar computer program for bovine control, based on their unique nasolabial plate imprints, the necessary data on production performance and the results for analysing and evaluation evaluating of the success of the agricultural business of the agricultural entity could be obtained. Furthermore, our calculation showed that the minimum number of minutiae points required for unique animal identification, with 0% possibility of repeating forms per point, was nine of 12 points. That makes the use of noseprint very interesting and applicable in animal identification systems, where monitoring of individual cows in real time ensures timely management decisions, resulting in improved large-scale farming (Zhang et al. 2023).

Given the easy obtainment of noseprints. such systems could be developed within milk parlours, or feedlots, where imprints can be taken on glossy photopaper and visualised with grey instant powder with very little level of practice, and then stored as digital images. Further work is very similar to everyday police work in fingerprint detection, as imprints images are examined under the AFIS magnifier. Simple muzzle examination by AFIS would allow for accurate ID chart creation and storage in a special database with animal IDs. In such a collection, as mentioned by Cihan et al. (2025), data on animal ownership, genetics, health and productivity status, history, and vaccination could be easily accessible. Based on the program created within the scope of this study, a simple comparison of nine of 12 points could differentiate for accurate ID confirmation. These images could be shared via mobile applications connected to the AFIS and muzzle imprint collection, enabling swift data flow that could lead to more efficient animal welfare, management and traceability between farms, veterinarians, and government institutions. Therefore, the foundation is already present and serves as a first step in the future development of biometric animal identification systems.

5 | Conclusion

In this study, we propose a cattle identification system based on forensic methods for trace detection. Muzzle imprints were taken on different surfaces and then visualised with different methods. The noseprints taken with glossy black photopaper and grey instant powder proved to be the best method for cow identification based on the nasolabial plate imprint. After digitisation of the visualised muzzle print, the image was analysed with an AFIS magnifier to detect specific beads and ridges at 12 selected minutiae points, thereby creating a unique animal ID chart. Computer simulation revealed that the minimum number of points required for accurate animal identification ID was nine of 12 points, with a 0% possibility of repetition in the points. Due to easy processing of images, it is possible to use this method in farm conditions and store the image of the imprint. Further development would require mobile data applications, connected to the AFIS analysed muzzle imprints collection, for swift data flow that could lead to more efficient animal welfare, management and traceability between farms, veterinarians, and government institutions. As a biometric method, analysing muzzle imprints with AFIS has the potential to be stored on large scale and used between countries, enabling transport control that is not susceptible to the problems faced by biometric methods.

Author Contributions

Daniel Špoljarić: conceptualisation; investigation; methodology; writing–original draft; writing–review and editing. **Luka Pajurin**: investigation; methodology. **Megi Kujundžić**: methodology; investigation. **Maja Ferenčaković**: formal analysis; data curation; software. **Anja Vrbaški**: formal analysis. **Branimira Špoljarić**: investigation; writing–original draft; writing–review and editing; supervision. **Gordan Mršić**: conceptualisation; methodology; investigation. **Mirela Pavić Vulinović**: investigation; writing–review and editing. **Marko Samardžija**: writing–original draft; writing–review and editing. **Maja Popović**: conceptualisation; resources; writing–original draft; writing–review and editing; supervision. **Silvijo Vince**: formal analysis; data curation; software; visualisation; writing–review and editing; supervision.

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Ethics Statement

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to, and the appropriate ethical review committee approval has been received (approval from the Ethics Committee for Animal Experimentation, Faculty of Veterinary Medicine, University of Zagreb, Croatia; records No.: 640-01/14-17/15; file No.: 251-61-01/139-14-1).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the supplementary material of this article.

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APPENDIX 1

6.1SAS 9.4 Code Used for Analyses of the Number of Possible Matching While Simultaneously Decreasing the Number of Points Used, Based on 100 Replications From Our Data Set

6.1.1A.1.1. Code for automatic factorial calculation

%let p = 1/6;

%let n = 12; **data** PROBABILITY; do k = **1** to &n; format probability **16.14**; probability_for_dot = pdf("Binomial", k, &p, &n); probability = probability_for_dot; output; end; **run**;

A.1.2. Macro for calculation of number of matches while decreasing the number of points, based on 100 replications from the dataset

%macro *rep*;

%local j;

%do j = 1 %to 100;

%macro *dots*;

%local i; %do i = 11 %to 11; ('number of chosen dots') proc surveyselect data = #name method = srs n = &i out = Sample_&i._&j; samplingunit _NAME_; run; proc sort data = sample_&i._&j; by cid; run; proc freq data = sample_&i._&j; tables col1*id/ outpct outexpect chisq out = tbl_&i._&j; by cid; run; data tbl_&i._&j; set tbl_&i._&j; rep = &j; run; %end;; %mend tocke; *%tocke* *run; %end;; %mend rep; *%rep* run; %macro *spoji11*; data dots_11; set %do i = 1 %to 100; tbl_11_&i %end; ; run; %mend; *%spoji11*;

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